PROJECT REPORT

Global Olympic Sports Performance Analysis

A Tableau-Based Analysis of Olympic Athlete Participation, Sports and Medal Patterns

Submitted by	Harsh Arjariya
Project Type	Individual Project
Institution	Medicaps University
Programme	B.Tech — Computer Science and Engineering
Project Duration	October 2026
Faculty / Guide	Not provided

Prepared using the supplied Project Report Format. Unknown measurements and unavailable evidence are identified explicitly.

Report Structure

Section	Contents
1. Introduction	Project Overview; Purpose
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Project facts used in this report

66	230	135,571
Distinct sports	Distinct NOC codes	Distinct athlete IDs

These values reflect the KPI calculations reported for the current Tableau workbook. 230 is a count of distinct NOC codes, not a verified count of modern sovereign countries.

1. INTRODUCTION

1.1 Project Overview

Global Olympic Sports Performance Analysis is an individual data-visualization project developed in Tableau. It uses two CSV sources—**athlete_events.csv** and **noc_regions.csv**—to explore Olympic athlete-event records, participation, sports categories, National Olympic Committee (NOC) codes, medal categories, and changes across Olympic years. The workbook includes KPI sheets, country and medal views, year-based charts, sport comparisons, two dashboards, and a Tableau Story with three story points.

1.2 Purpose

The purpose is to turn Olympic event data into understandable visual summaries so a viewer can explore participation and medal patterns across countries/NOC codes, sports, and years. The dashboards are intended to support exploratory analysis and presentation rather than to establish causal explanations for why a country or sport performs well.

Project scope and data caveats

• Known KPI calculations: COUNTD([Sport]) = 66; COUNTD([NOC]) = 230; COUNTD([ID]) = 135,571.

• Known dashboard filters: Team and Medal.

• The available sources do not include a sports-investment field; investment-versus-medal analysis is outside the current implemented scope.

• A regional pie chart based on record/participation distribution must not be described as medal efficiency.

• The City field alone does not validate a host-country medal comparison; that would require a verified host-city-to-country mapping and a defined medal measure.

2. IDEATION PHASE

2.1 Problem Statement

Olympic athlete-event datasets can be difficult to interpret as raw tables because they contain many records across athletes, sports, events, years, countries/NOC codes, seasons, and medal categories. A viewer may struggle to identify high-level patterns without a structured set of summaries and interactive charts. The project addresses this information-discovery problem through Tableau dashboards and a guided Story.

2.2 Empathy Map Canvas

Perspective	Project interpretation
Says	“I want to understand Olympic participation and medal patterns without manually scanning a large CSV.”
Thinks	Which sports, countries/NOC codes, and years stand out in the available data? What do the displayed measures actually count?
Does	Opens dashboards, checks KPI cards, compares country and sport charts, and uses Team and Medal filters.
Feels	Needs a clear, readable dashboard and confidence that labels and calculations are not misleading.
Needs / pains	Consistent labels, quick comparisons, visible definitions, and clear caveats where data is missing or a measure is only a proxy.

2.3 Brainstorming

• Use KPI cards for distinct sports, distinct NOC codes, and distinct athlete IDs.

• Compare medal records across country/NOC categories and highlight leading entries in the displayed ranking.

• Use year-based charts to explore participation and medal-record trends over time.

• Use sport-based charts to compare categories, while making the aggregation and denominator explicit.

• Add filters and a three-point Story to guide viewers through the main analytical views.

• Defer investment and true medal-efficiency analysis until suitable source fields and definitions are available.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Viewer activity	Design response
Open	Opens Dashboard 1 and looks for an overall summary.	KPI cards provide an initial orientation.
Explore	Reviews country, medal, year, and sport charts.	Charts provide multiple perspectives on the available data.
Filter	Selects Team or Medal to narrow the view.	Known dashboard filters support focused exploration.
Compare	Moves between Dashboard 1, Dashboard 2, and Story points.	Separate dashboards and Story points organize the analysis.
Interpret	Forms observations from the displayed measures.	Labels and caveats clarify what is counted and what is not available.

3.2 Solution Requirement

Functional requirements

• Connect the two CSV sources in Tableau and expose relevant fields such as ID, Sport, NOC, Year, Medal, Team, Season, City, and region mapping where available.

• Calculate distinct counts for Sport, NOC, and ID.

• Present country/medal, year, and sport views using suitable chart types.

• Provide two dashboards and a Story with three story points.

• Allow viewers to use the known Team and Medal filters.

Non-functional requirements

• Readable labels, consistent colour usage, clear titles, and understandable metric definitions.

• Correct aggregation and honest reporting of missing or unverified values.

• A layout suitable for a live demonstration and presentation.

3.3 Data Flow Diagram

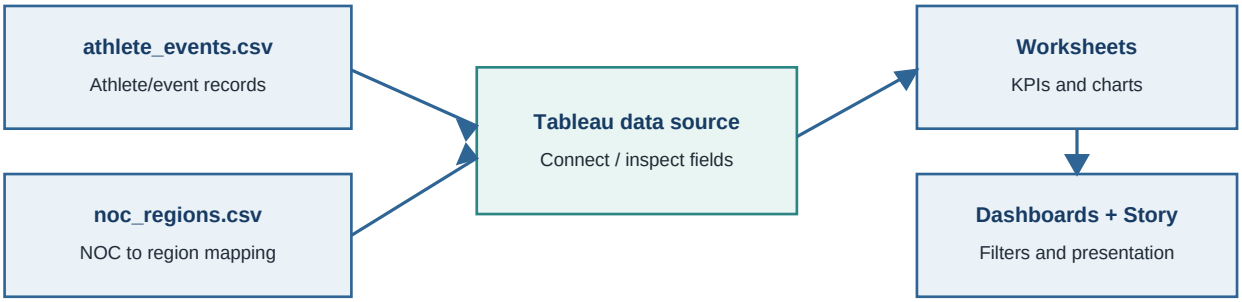


Figure 1. Conceptual data flow based on the project workflow; it is not a database deployment diagram.

3.4 Technology Stack

Component	Use in the project
Tableau	Connect CSV files, create worksheets, calculated KPI measures, dashboards, filters, and Story points.
CSV files	athlete_events.csv contains athlete/event records; noc_regions.csv provides NOC-to-region mapping.
Tableau Public	Public dashboard link supplied by the student; a separate Story-specific URL has not been verified.
SQL / Snowflake / Streamlit	Not used in the described implementation.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Tableau is a suitable fit because the project goal is exploratory visual analysis rather than building a custom data-entry application. It supports connecting CSV sources, aggregating measures, comparing categories, adding filters, and assembling multiple views into dashboards and a guided Story.

4.2 Proposed Solution

The solution is a Tableau workbook organized around a small set of KPI cards and supporting visualizations. The dashboards combine country/NOC and medal comparisons with year- and sport-based analysis. Team and Medal filters let a viewer focus on subsets of the available data. The three-point Story provides a presentation path through the key views.

4.3 Solution Architecture

Layer	Description
Input	Two CSV sources: athlete_events.csv and noc_regions.csv.
Preparation	Connect the sources in Tableau; inspect relevant dimensions and measures; use NOC/region mapping where available. Specific null/duplicate cleaning results were not supplied and remain unverified.
Analysis	Create distinct-count KPI calculations and aggregate views by NOC/region, sport, year, and medal category.
Presentation	Worksheets are arranged into two dashboards; a Tableau Story contains three story points.
Publication	Tableau Public dashboard URL supplied in Appendix. The supplied Story URL is identical to the Dashboard URL, so a distinct Story URL should be checked.

Known KPI calculations

KPI	Tableau calculation	Reported result
Total sports	COUNTD([Sport])	66
Distinct NOC codes	COUNTD([NOC])	230
Unique athlete IDs	COUNTD([ID])	135,571

Interpretation note: COUNTD([ID]) represents distinct athlete IDs in the workbook's calculation, not the number of event rows. COUNTD([NOC]) counts NOC codes and should not be labelled as an exact count of present-day sovereign countries.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was carried out as an individual project during October 2026. The schedule below is a retrospective summary of work areas, not a verified day-by-day time log.

Work area	Activities	Status based on available project history
Data connection	Connect athlete_events.csv and noc_regions.csv; inspect available fields.	Completed in Tableau workbook, based on project history.
KPI sheets	Create distinct-count KPI sheets for Sport, NOC, and ID.	Known results: 66, 230, and 135,571.
Visualization development	Build country/medal, year, sport, participation, and map-based views.	Multiple chart types are present; exact count of visualizations is not verified.
Dashboard assembly	Arrange worksheets and use Team/Medal filters.	Two dashboards are known to exist.
Story and review	Create three Story points and prepare a presentation narrative.	Three Story points are known; final screenshot and URL verification remain pending.

Planning assumptions and outstanding checks

• The exact start/end dates within October 2026 were not provided.

• The number of charts per dashboard has not been verified.

• Data row counts, file sizes, null counts, duplicate counts, and cleaning outcomes were not provided.

• No measured filter response times or query execution timings are available; the project uses Tableau and CSV files rather than Snowflake.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The testing summary below distinguishes features known to exist from measurements that have not been captured. This avoids reporting fabricated performance results.

Test area	Test / expected outcome	Current evidence / status
Data rendering	CSV sources connect and fields are available for analysis.	Source files are used in the Tableau workbook; exact file sizes and row counts are not verified.
KPI calculations	Distinct counts produce the documented KPI values.	Reported workbook values: 66 distinct Sport, 230 distinct NOC, 135,571 distinct ID.
Filters	Team and Medal filters narrow the displayed data as configured.	Filters are documented in the dashboard; formal response-time testing was not measured.
Dashboard display	Two dashboards display charts and KPI views.	Two dashboards are known to exist; exact visualization count is not verified.
Story navigation	Story contains three story points.	Known from project history; test each point in the final published workbook.
Data quality	Check nulls, duplicate event records, and NOC-region mapping.	Actual null/duplicate counts and cleaning outcomes were not supplied; mark as not verified.
Query history / execution time	Record the slowest database query if a database platform is used.	Not applicable to the described Tableau + CSV workflow; no Snowflake query history exists in the project details.

Recommended final manual checks

• Open both dashboards and verify all charts render correctly.

• Select Team and Medal filters and check that relevant views update.

• Verify KPI labels and aggregation definitions.

• Open all three Story points and confirm the narrative order.

• Check the Tableau Public link in a signed-out or private browser session if possible.

No numerical response-time, query-time, null-count, or duplicate-count result is claimed because those measurements were not provided.

7. RESULTS

7.1 Output Screenshots

Actual Tableau screenshots supplied by the student.

Figure 2. Dashboard 1 · KPI overview and country/year/sport views

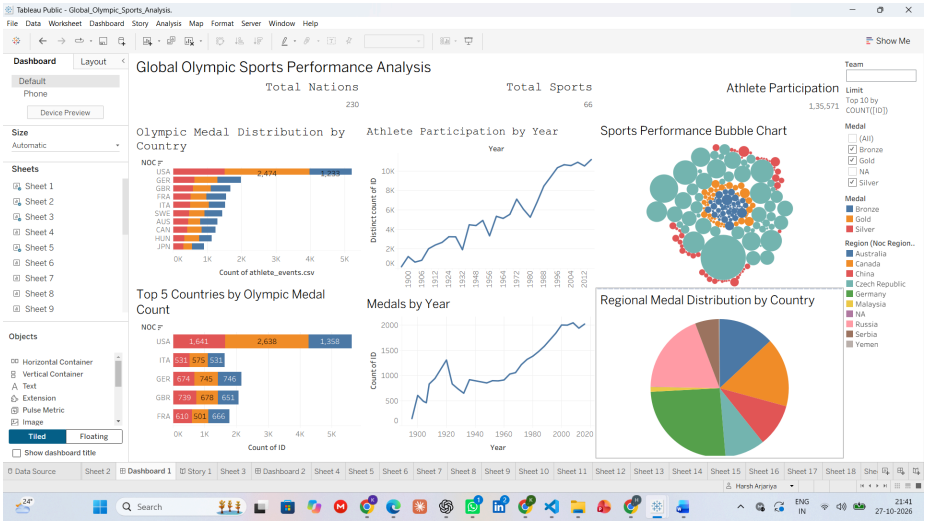
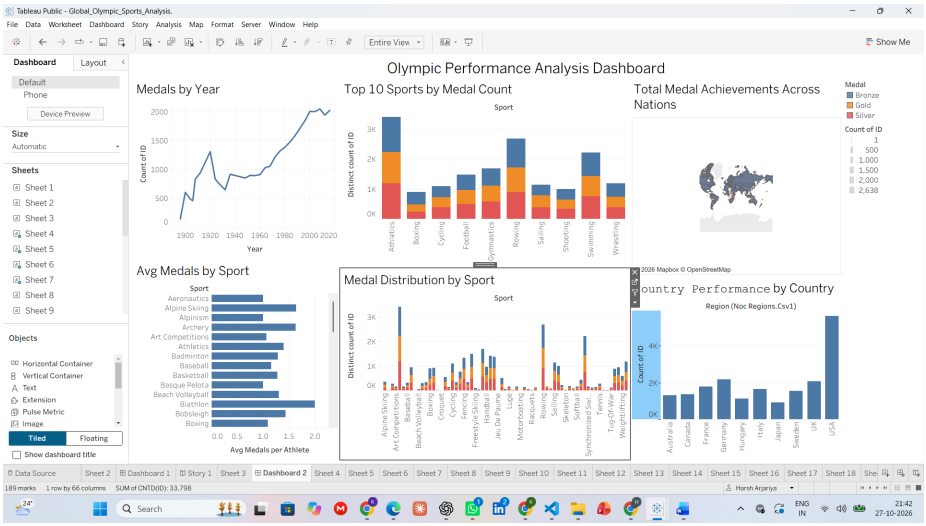


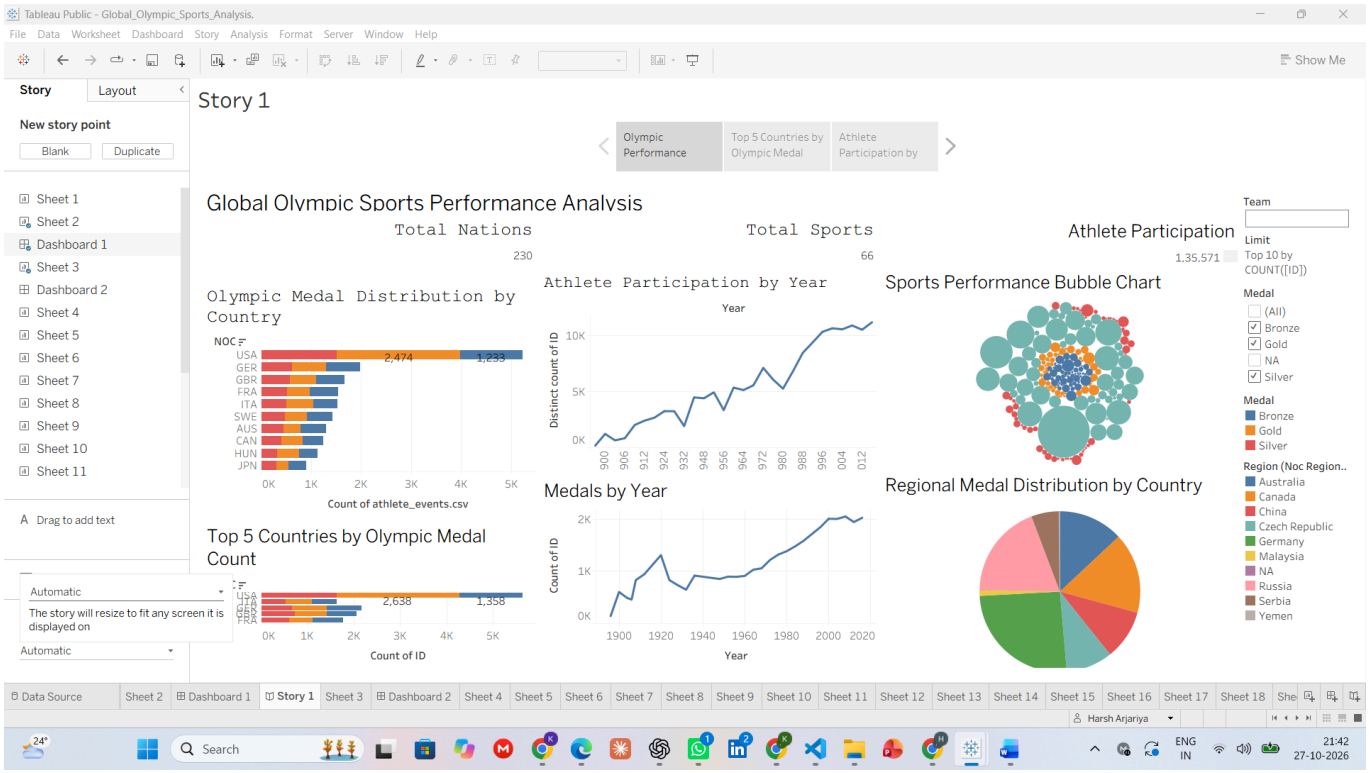
Figure 3. Dashboard 2 · medal trends, sports, map and country performance



Charts should be interpreted using each worksheet's aggregation and field definitions.

7. RESULTS (CONTINUED)

Figure 4. Tableau Story · three story points



The Story navigation shows three points: "Olympic Performance", "Top 5 Countries by Olympic Medal", and "Athlete Participation by Year". The screenshot documents the Story as it appeared in Tableau at the time of capture.

Note: Verify that the published Tableau Public Story URL points to this Story before submission.

8. ADVANTAGES & DISADVANTAGES

Advantages

• Transforms large tabular records into visual summaries that are easier to explore.

• KPI cards provide a quick overview of distinct sports, NOC codes, and athlete IDs.

• Country/medal, year, and sport views allow comparisons from different perspectives.

• Team and Medal filters support interactive exploration.

• Two dashboards and a three-point Story provide both exploratory and presentation-oriented views.

Disadvantages / limitations

• The reliability of insights depends on the quality, coverage, and field definitions of the CSV sources.

• Distinct NOC codes are not the same as a verified count of modern countries.

• The known sources do not include sports investment data, so investment-versus-medal analysis cannot be completed from them.

• The current regional pie chart represents record/participation distribution, not medal efficiency.

• A host-country comparison requires a verified host-city-to-country mapping and a defined medal measure.

• Formal data-quality counts and performance timing measurements have not been supplied.

The Story navigation shows three points: "Olympic Performance", "Top 5 Countries by Olympic Medal", and "Athlete Participation by Year". The screenshot documents the Story as it appeared in Tableau at the time of capture.

9. CONCLUSION

Global Olympic Sports Performance Analysis demonstrates how Tableau can be used to explore Olympic athlete-event data through KPI cards, country and medal comparisons, time-based views, sport comparisons, dashboards, filters, and a guided Story. The workbook reports 66 distinct sports, 230 distinct NOC codes, and 135,571 distinct athlete IDs using its current KPI calculations.

The project provides a practical example of converting CSV data into an interactive analytical presentation. Its conclusions should be limited to what the available fields and visible charts support. The report therefore separates known workbook facts from items that still require validation, including data-quality checks, exact chart counts, and screenshot evidence.

The current implementation uses Tableau and CSV files only. It does not claim a Snowflake pipeline, investment data analysis, or a validated host-country medal comparison.

10. FUTURE SCOPE

• Add a reliable sports investment or funding dataset and a clear definition of investment-versus-medal performance.

• Add a verified Olympic host-city-to-country mapping to support valid host-country medal comparisons.

• Define a transparent medal-efficiency formula, including its denominator, before calculating regional efficiency percentages.

• Add data-quality checks and document actual null, duplicate, and mapping-validation results.

• Capture dashboard and Story screenshots and verify the distinct public URL for the Story.

• Count visualizations per dashboard and document the final workbook inventory.

• Test Team and Medal filter behavior systematically and record response observations if needed.

• Improve accessibility through consistent titles, readable labels, colour choices, and explanatory tooltips.

11. APPENDIX

11.1 Source Code (if any)

This is a Tableau workbook project. No standalone source-code repository or code files were provided for this report. The key calculations documented from the workbook are:

Total Sports
COUNTD([Sport])

Distinct NOC Codes
COUNTD([NOC])

Unique Athlete IDs
COUNTD([ID])

11.2 Dataset Link

The project uses the local files **athlete_events.csv** and **noc_regions.csv**. A public dataset URL was not supplied in the project details; add the dataset source URL used for download if required by the instructor.

11.3 GitHub & Project Demo Link

Tableau Public Dashboard:
https://public.tableau.com/app/profile/harsh.arjariya/viz/Global_Olympic_Sports_Analysis_/Dashboard1?publish=yes

Tableau Story URL supplied:
https://public.tableau.com/app/profile/harsh.arjariya/viz/Global_Olympic_Sports_Analysis_/Dashboard1?publish=yes

The Dashboard and Story URLs supplied are identical. A separate Story-specific URL has not been verified; replace the Story link above if the published Story has its own URL.

GitHub: Not provided — add repository URL if one exists.
Faculty / Guide: Not provided.
Team ID: PNT2022TMIDxxxxxx (template placeholder retained where relevant).

Report evidence note

File sizes, row counts, data-quality counts, cleaning outcomes, exact chart counts, filter response times, query execution timings, and actual screenshot files were not supplied for this report. These items are not fabricated and should be updated if verified evidence becomes available.

End of Report

SUPPLEMENT: TEMPLATE-ALIGNED PROJECT DETAILS

This supplement adds explicit items reflected in the supplied 11-page report format. It complements the main report and preserves the actual project scope: Tableau Public with two CSV files.

A. Dataset Description and Data Collection

The project uses two local CSV files connected in Tableau Public. The first contains historical Olympic athlete-event records. The second provides mapping information between NOC codes and region names. Exact source download URL, file size, and row counts were not provided and are therefore not stated as verified facts.

Dataset / field	Description used in the project
athlete_events.csv	Main Olympic athlete/event records used for sport, athlete, team, year, event, and medal analysis.
noc_regions.csv	Supporting NOC-code-to-region mapping used to provide readable geographic labels.
Name, Age, Sex	Athlete attributes available in the Olympic athlete-event dataset.
Team, NOC	Team represented and National Olympic Committee code.
Year, Season, City	Olympic edition and event context used for time and category exploration.
Sport, Event	Sport category and specific event.
Medal	Medal category (Gold, Silver, Bronze); NA indicates no medal recorded in the source. Medal charts should filter to medal categories when measuring awarded medals.

Data collection method

The CSV files were loaded into Tableau Public and used as the basis for the workbook's worksheets, dashboards, and Story. No separate database, Snowflake warehouse, Streamlit application, or SQL pipeline is part of the described implementation.

B. Data Preparation — Explicit Steps

Step	Action / scope
1. Load data	Connect athlete_events.csv in Tableau Public and access the NOC-region mapping file.
2. Review fields	Inspect fields used for analysis, including Name, Age, Sex, Team, NOC, Year, Season, City, Sport, Event, and Medal where present.
3. Connect geographic labels	Use NOC and region information to support country/region labels in geographic and country-oriented views.
4. Filter medal views	Use Gold, Silver, and Bronze for medal-focused visualizations; exclude NA medal records from medal counts where configured.
5. Prepare visual fields	Use country/NOC, medal, year, sport, and athlete ID fields in Tableau visualizations.
Validation status	Specific null counts, duplicate counts, and detailed cleaning outcomes were not supplied; these checks remain unverified rather than being claimed as completed.

C. Data Visualizations Created

Visualization	Chart type and purpose
Olympic Medal Distribution by Country	Stacked horizontal bar chart comparing Gold, Silver, and Bronze records across NOC/country labels.
Top 5 Countries by Olympic Medal Count	Stacked horizontal bar chart comparing the five selected leading country/NOC categories.
Athlete Participation by Year	Line chart showing the distinct athlete-ID count by Olympic year as configured in the workbook.
Medals by Year	Line chart using the workbook's current count measure across Olympic years; verify the measure label before describing it as medal totals.
Top 10 Sports by Medal Count	Stacked bar chart comparing medal-category counts for selected sports.
Average Medals by Sport	Horizontal bar chart for the workbook's average-medal measure; interpret according to its actual aggregation.
Sports Performance Bubble Chart	Bubble chart with Sport categories, distinct athlete IDs as size, and medal category as colour.
Total Medal Achievements Across Nations	World map showing geographic distribution using the workbook's available country/region mapping and measure.
Country Performance by Country	Bar chart comparing the selected country/region categories using the workbook's displayed count measure.
Regional distribution pie chart	Shows record/participation distribution by region in the current workbook; it is not a validated medal-efficiency percentage.

D. Dashboard Design and Interactivity

Dashboard 1 contains the KPI cards for distinct NOC codes, distinct sports, and distinct athlete IDs, alongside country medal distribution, athlete participation by year, the sports bubble chart, top-country comparison, medals by year, and a regional distribution pie chart. Team and Medal filters are visible in the supplied screenshot.

Dashboard 2 contains medals by year, top sports by medal count, a world map, average medals by sport, medal distribution by sport, and country performance by country. The supplied screenshot shows a floating chart overlapping part of the lower dashboard; adjust this layout before final presentation if possible.

Known KPI calculations

KPI	Tableau calculation	Reported value
Total Sports	COUNTD([Sport])	66
Distinct NOC Codes	COUNTD([NOC])	230
Unique Athlete IDs	COUNTD([ID])	135,571

Label note: use “Distinct NOC Codes” instead of “Total Nations” if you want the KPI label to match COUNTD([NOC]) precisely. NOC codes do not necessarily equal the number of present-day sovereign countries.

E. Functional Testing and Results

Test	Expected / reviewed behaviour	Evidence status
Chart display	Dashboard charts render and show the selected data.	Screenshots supplied for Dashboard 1 and Dashboard 2.
Medal filter	Gold, Silver, and Bronze can be selected; NA is excluded from medal analysis where configured.	Filter is visible in the supplied Dashboard 1 screenshot.
Team filter	Viewer can focus on a selected team.	Filter control is visible in the supplied Dashboard 1 screenshot.
Dashboard layout	Charts and KPI cards remain readable and do not unintentionally overlap.	Dashboard 2 screenshot shows a floating chart overlapping other content; improve before submission.
Story navigation	Story has three story points: Olympic Performance; Top 5 Countries by Olympic Medal; Athlete Participation by Year.	Story screenshot supplied.
Performance timing	Filter response times or database query execution time are measured.	Not measured / not applicable to the current Tableau + CSV workflow; no timing values are claimed.

Results summary

The screenshots document two dashboards and a three-point Tableau Story. The workbook reports 66 distinct sports, 230 distinct NOC codes, and 135,571 distinct athlete IDs. The country and sport charts provide comparative views; trends should be described from the visible axes and measures, not as causal explanations.

F. Conclusion and Future Scope — Additional Points

The project demonstrates how Tableau Public can turn Olympic CSV data into visual summaries for country/medal comparisons, sports comparisons, and year-based exploration. It also provides practice in connecting data, defining measures, arranging dashboard components, and using filters.

Potential improvements aligned with the supplied report format include adding season-wise comparisons, more sports-focused views, additional filters, verified data-quality checks, and improved dashboard layout. Investment-versus-medal analysis requires a separate investment dataset. Host-country medal comparisons require a verified host-city-to-country mapping. A medal-efficiency percentage requires an explicit formula and denominator.

G. References

Reference	Use in this report
Tableau Public — https://public.tableau.com/	Tool used to create and publish the workbook.
Olympic Athlete Events Dataset	Historical athlete-event, sport, year, team, and medal records from athlete_events.csv. The precise original download URL was not supplied.
NOC Regions Dataset	NOC-code and region-name mapping from noc_regions.csv. The precise original download URL was not supplied.
Student-provided Tableau screenshots	Screenshots of Dashboard 1, Dashboard 2, and Story supplied for the Results section.

Project links

Dashboard:
https://public.tableau.com/app/profile/harsh.arjariya/viz/Global_Olympic_Sports_Analysis_/Dashboard1?publish=yes

Story: Same URL was supplied by the student; a separate Story-specific URL remains to be verified. **GitHub:** Not provided.

End of supplement